

# Advancing the Sustainability of Controlled Environment Agricultural Systems with Internet of Things (IoT) and Artificial Intelligence



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**Penn State Ag Progress Days**

**August 14, 2024 – Penn State Russell E. Larson Agricultural Research Center**

# Acknowledgements



Project:

## Advancing the Sustainability of Indoor Urban Agricultural Systems

Funding source:

2023 Pennsylvania Federal Specialty Crop Block Grant Program (Grant Agreement No.: C940001528)



Department of Plant Science, Department of Agricultural and Biological Engineering, & Department of Food Science



# What is Indoor Agriculture and why it matters?



**Greenhouse systems**



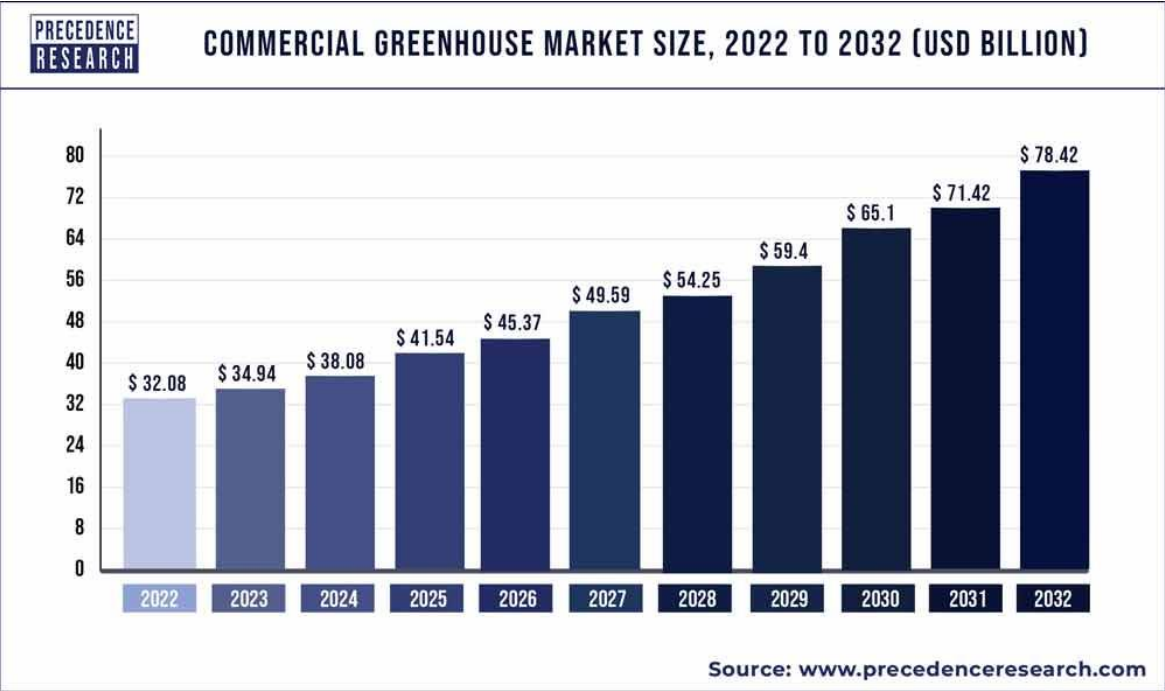
**Vertical farming systems**



# What is Indoor Agriculture and why it matters?

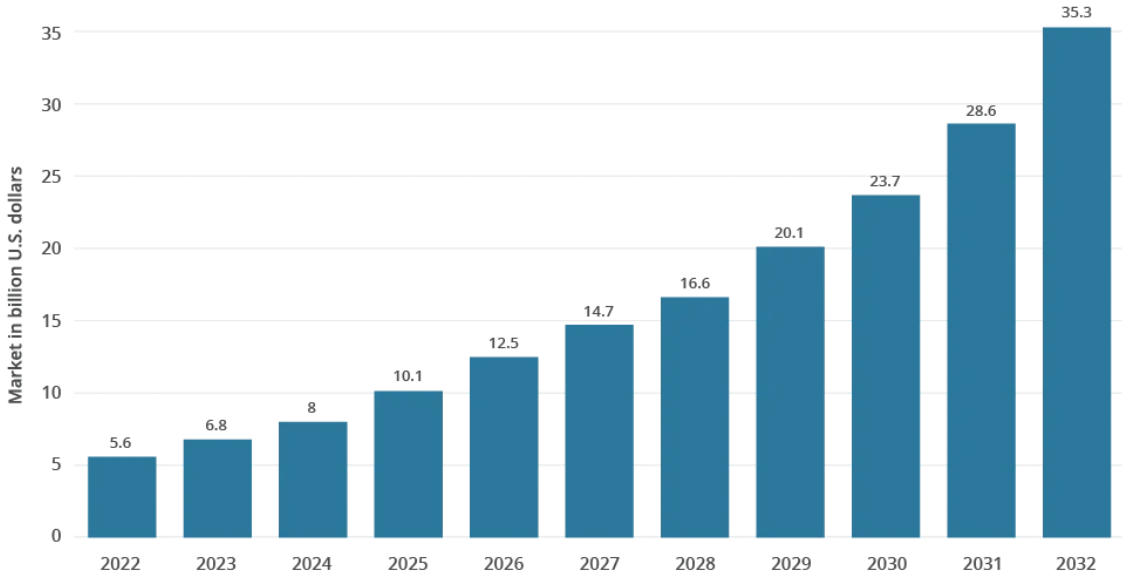


Estimated growth for the Global Greenhouse market



USD 32.08 billion in 2022

Estimated growth for Global Vertical Farming market

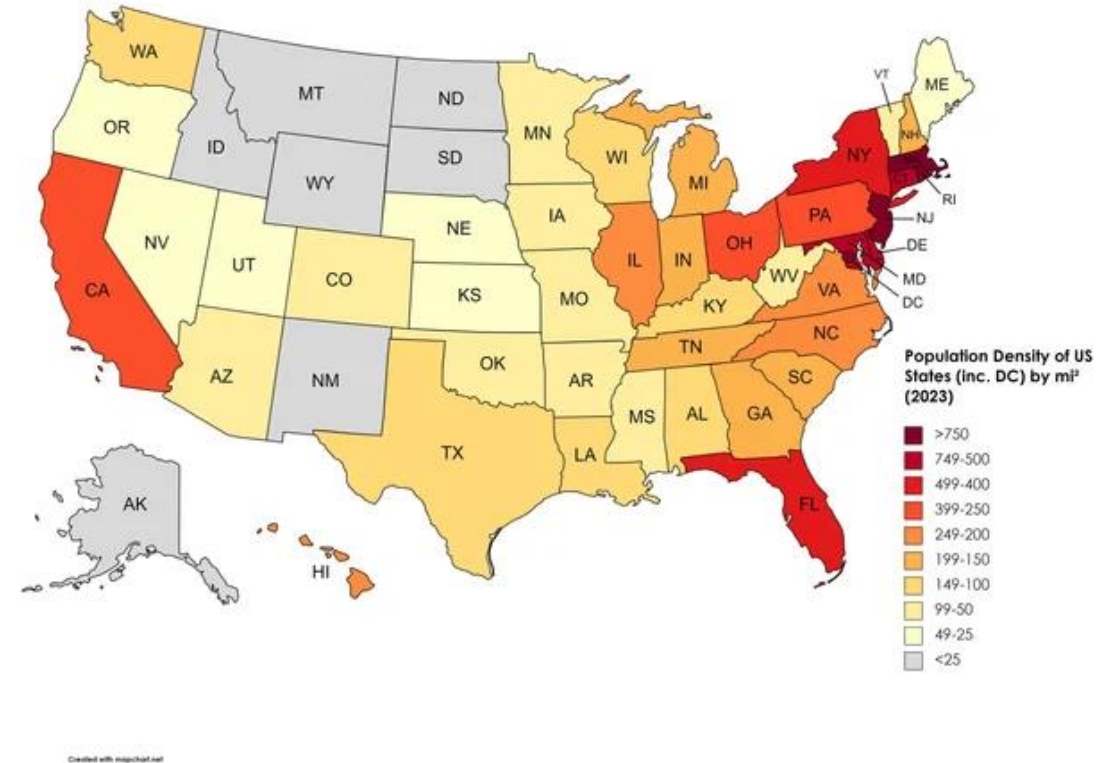


USD 5.6 billion in 2022



# What is Indoor Agriculture and why it matters?

- The U.S. Northeast region includes about **22% of the U.S. population**, but **only 3% of the nation's cropland**
- The cultivation of vegetables within **the U.S. Northeast accounts for only 26% of vegetables consumed in the region**
- Currently, **food and nutrition insecurity affects 7 million people throughout the Northeast U.S.**
- Investing in advanced growing systems to produce high-quality fresh and nutrient-dense vegetables is a key priority for enhancing the food and nutrition security of the region.



# What is Indoor Agriculture and why it matters?

## Bright Farms

Selinsgrove, PA

4 acres  
high-tech  
hydroponic  
greenhouse

\$20 million  
investment





# What is Indoor Agriculture and why it matters?

## Little Leaf Farms

Macadoo, PA

10 acres  
high-tech  
hydroponic  
greenhouse

\$300 million  
investment





# What is Indoor Agriculture and why it matters?



Small scale commercial indoor farming operations growing especially around urban areas

**Department of Plant Science & Department of Agricultural and Biological Engineering**



# What is the efficiency of indoor farming systems?

## Research questions:

- 1) What is the efficiency of alternative soilless systems?
- 2) Can IoT and Artificial Intelligence technology help improve the efficiency and sustainability of indoor farming systems?

|                                                                                     |                                   | Total landed cost<br>(USD/kg) | Annual yield<br>(kg/sq.m) | Water use<br>(l/kg) | Electricity use<br>(KWh/kg) |
|-------------------------------------------------------------------------------------|-----------------------------------|-------------------------------|---------------------------|---------------------|-----------------------------|
|    | Leaf greens<br>(Vertical farm)    | 3x                            | 35x                       | -90%                | 75x                         |
|    | Peppers<br>(High-tech greenhouse) | 1.5x                          | 10x                       | -85%                | 30x                         |
|  | Strawberries<br>(Vertical farm)   | 3x                            | 7x                        | -50%                | 120x                        |

Source: Infopulse



# What is the efficiency of alternative soilless systems?

Winter of 2024

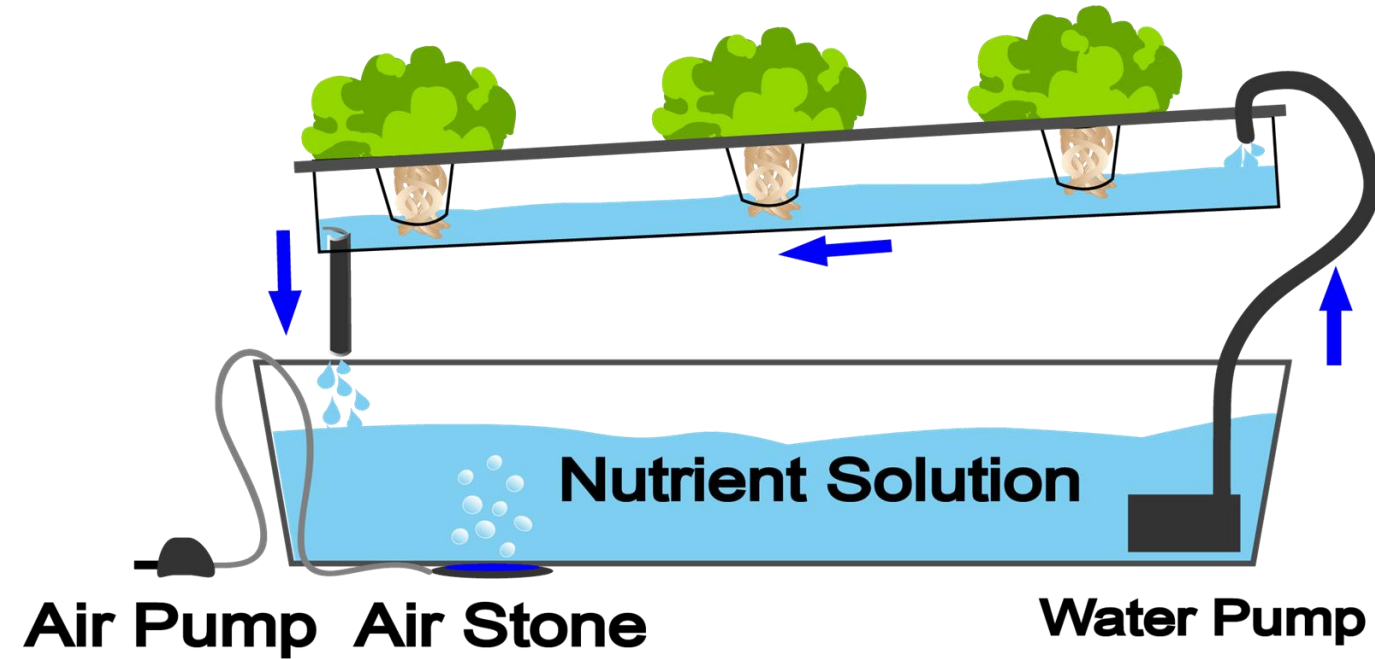
Side-by-side  
comparison of  
five alternative  
soilless growing  
systems using  
 $1.2 \times 1.2$  m  
modules





# What is the efficiency of alternative soilless systems?

## Nutrient Film Technique

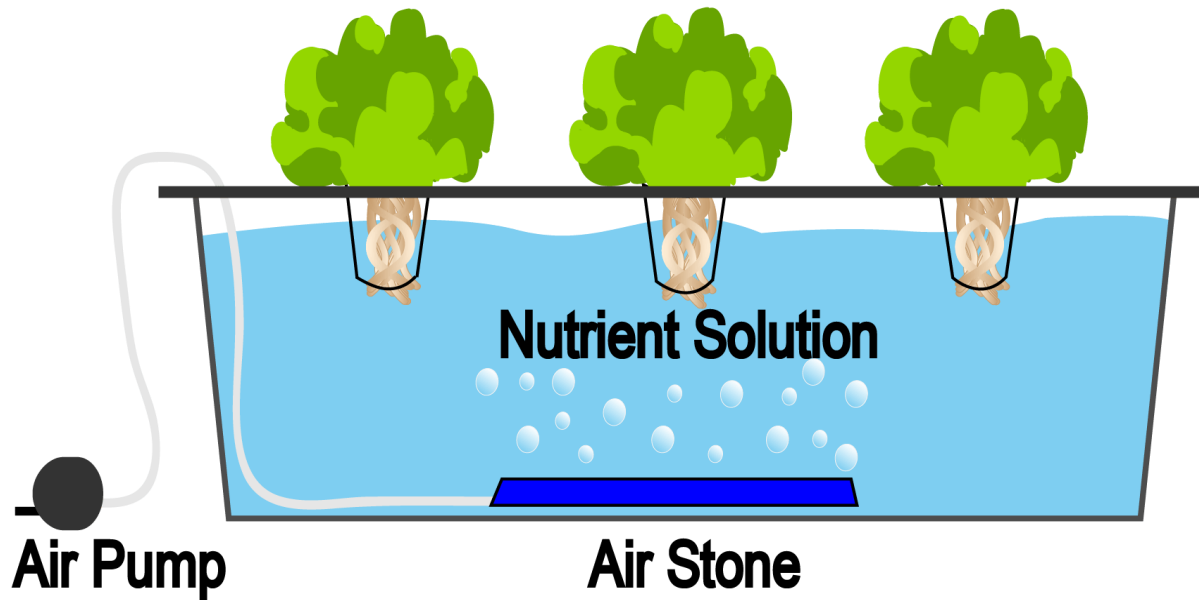


Credit: [nosoilsolution.com](http://nosoilsolution.com)



# What is the efficiency of alternative soilless systems?

## Deep Water Culture (DWC)

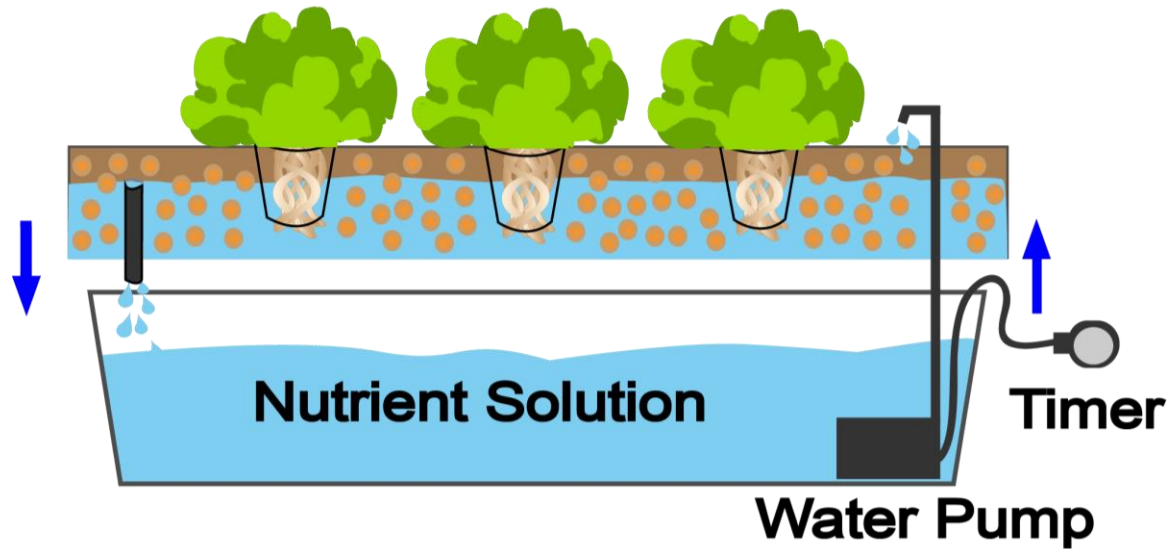


Credit: [nosoilsolution.com](http://nosoilsolution.com)



# What is the efficiency of alternative soilless systems?

## Subirrigation or Ebb and Flow



Credit: [nosoilsolution.com](http://nosoilsolution.com)





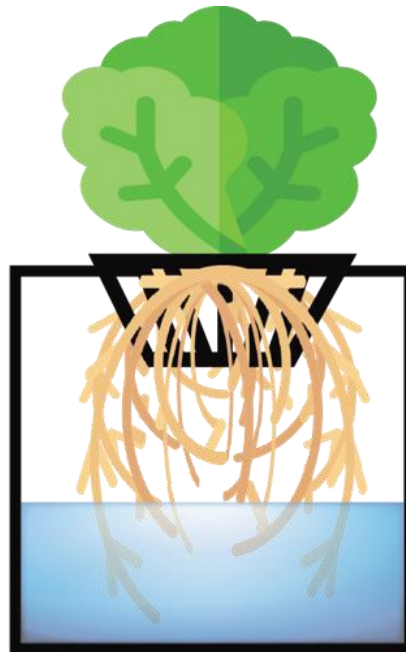
# What is the efficiency of alternative soilless systems?

## Kratky or Water gap

### Kratky Method Of Hydroponics



As the plant grows the water level drops, providing the roots with more oxygen.

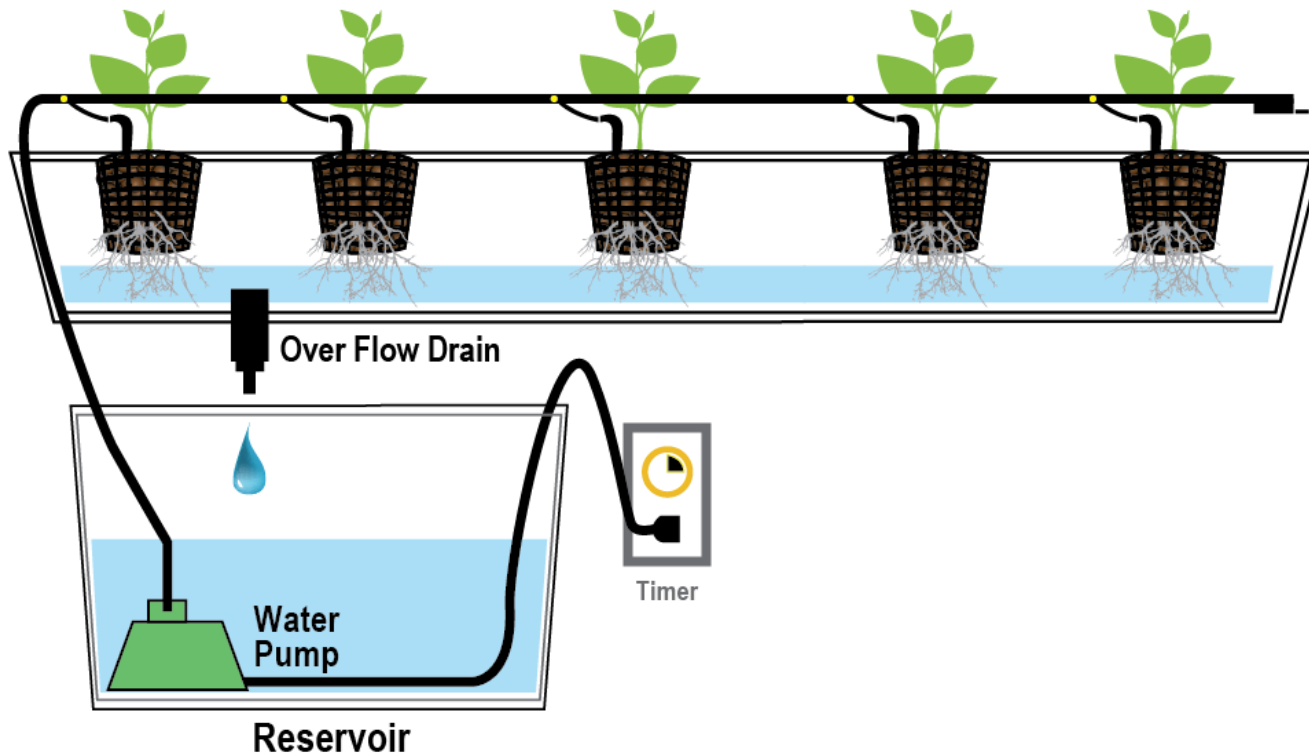


Credit: [nosoilsolution.com](http://nosoilsolution.com)



# What is the efficiency of alternative soilless systems?

## Drip system



Credit: [nosoilsolution.com](http://nosoilsolution.com)



# What is the efficiency of alternative soilless systems?

Test crop: Baby bok choy  
(*Brassica rapa subsp. chinensis*)

Density: 25 plant/m<sup>2</sup> (2.3 plant/ft<sup>2</sup>)

Standard nutrient solution

Experimental design:  
RCBD with three replicates

Planting: 2/9/2024

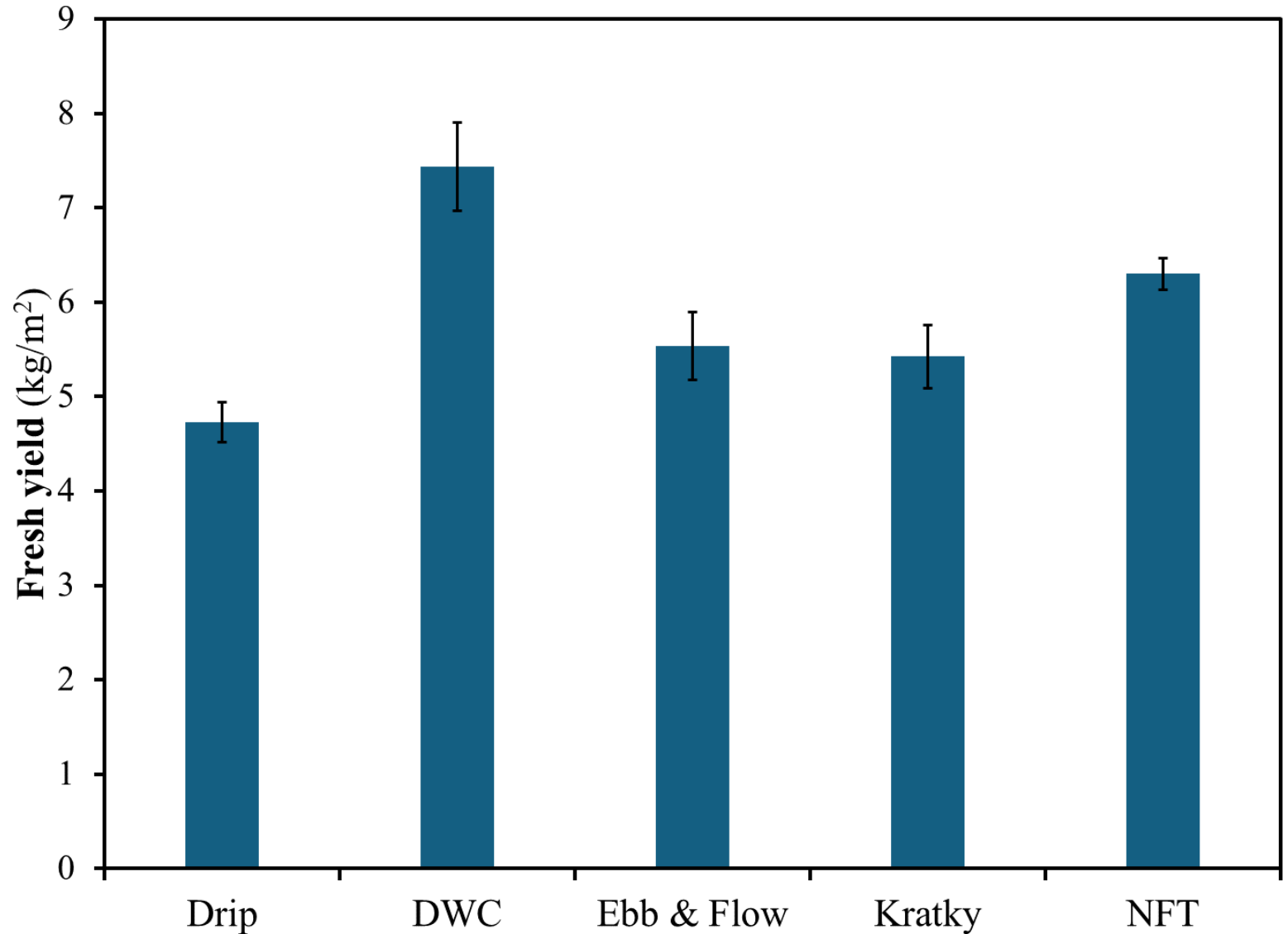
Harvest: 3/15/2024 (35 DAP)





# What is the efficiency of alternative soilless systems?

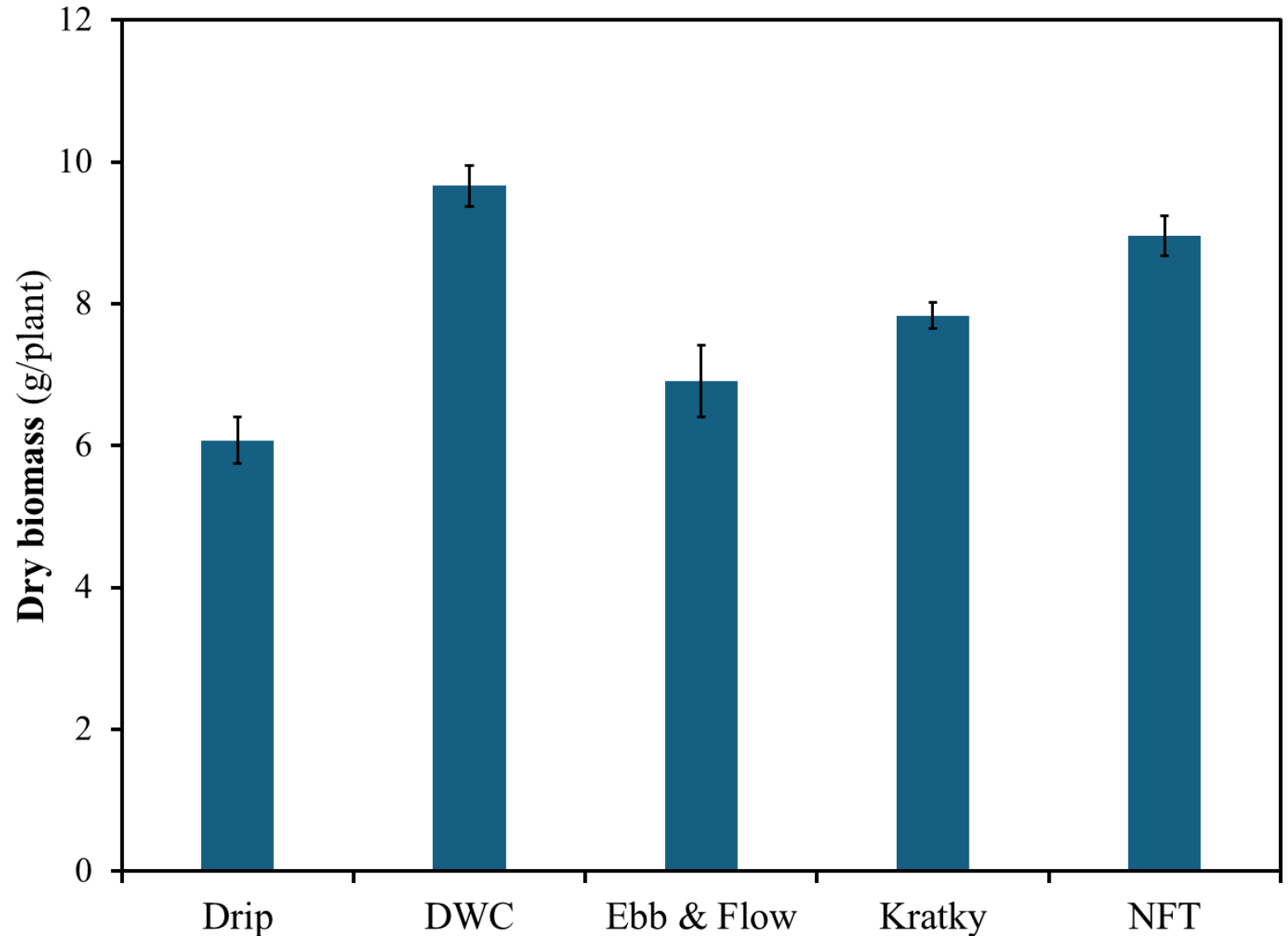
## Impact on Fresh Yield





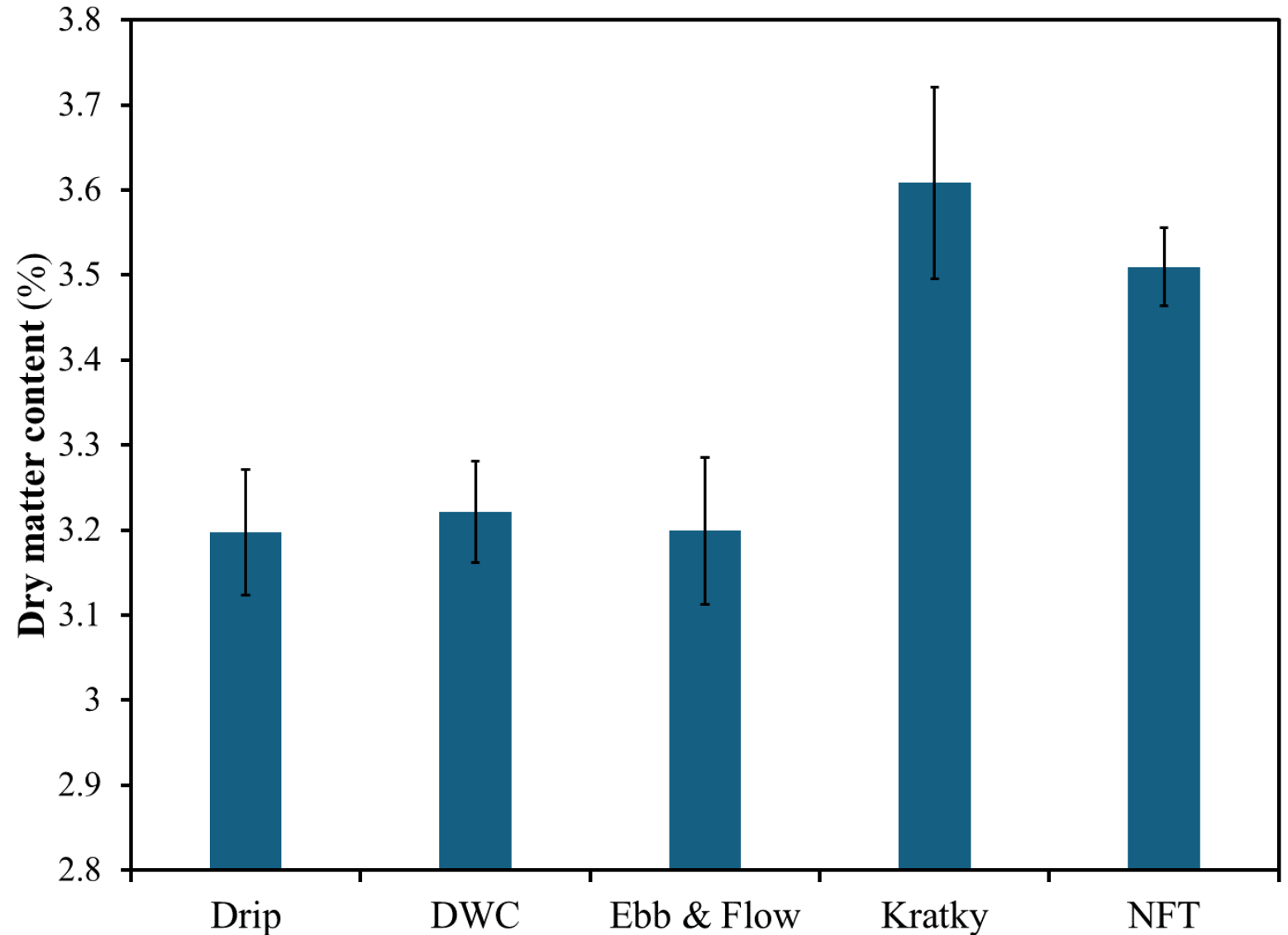
# What is the efficiency of alternative soilless systems?

## Impact on dry biomass



# What is the efficiency of alternative soilless systems?

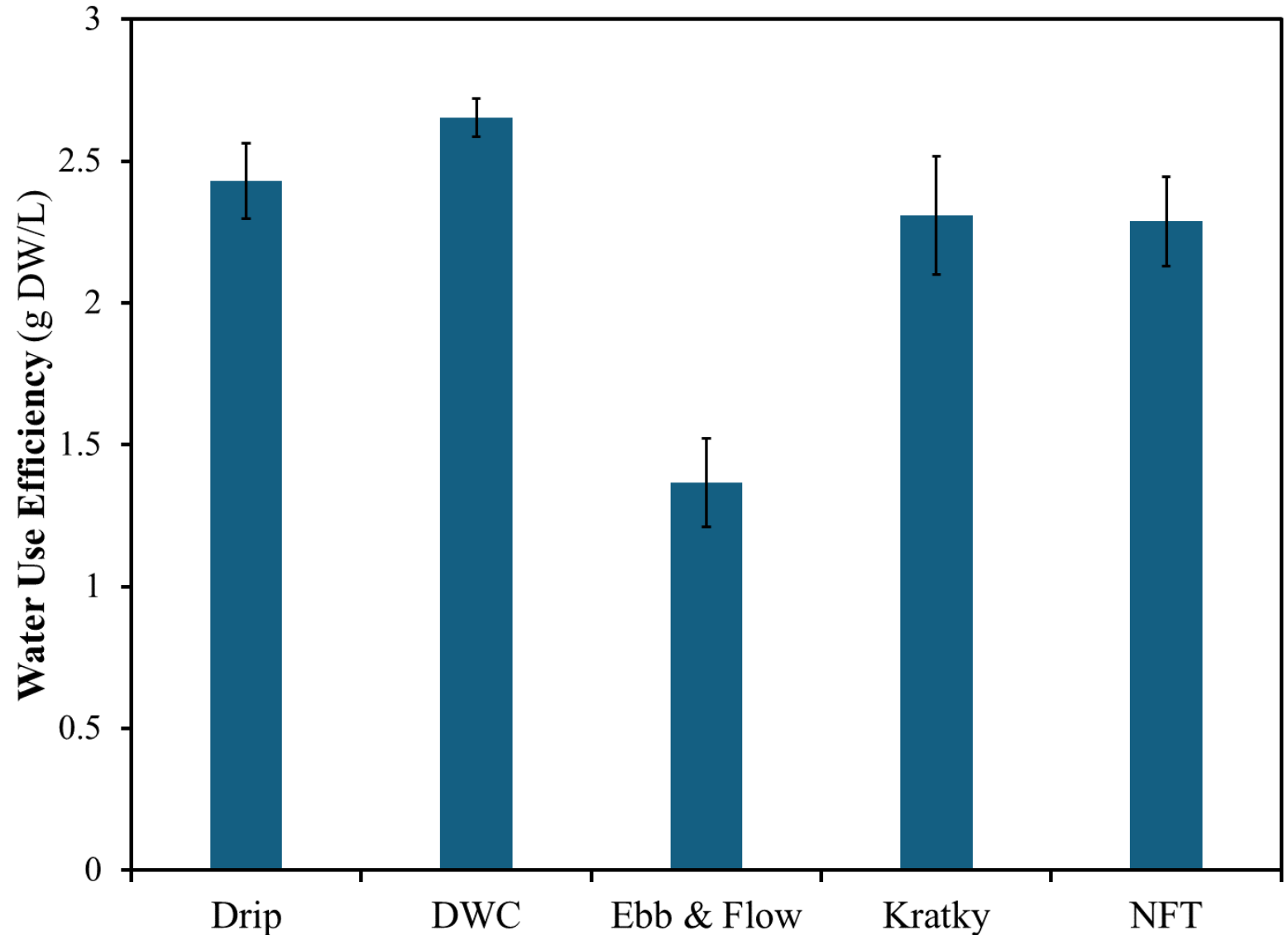
Impact on  
dry matter  
content





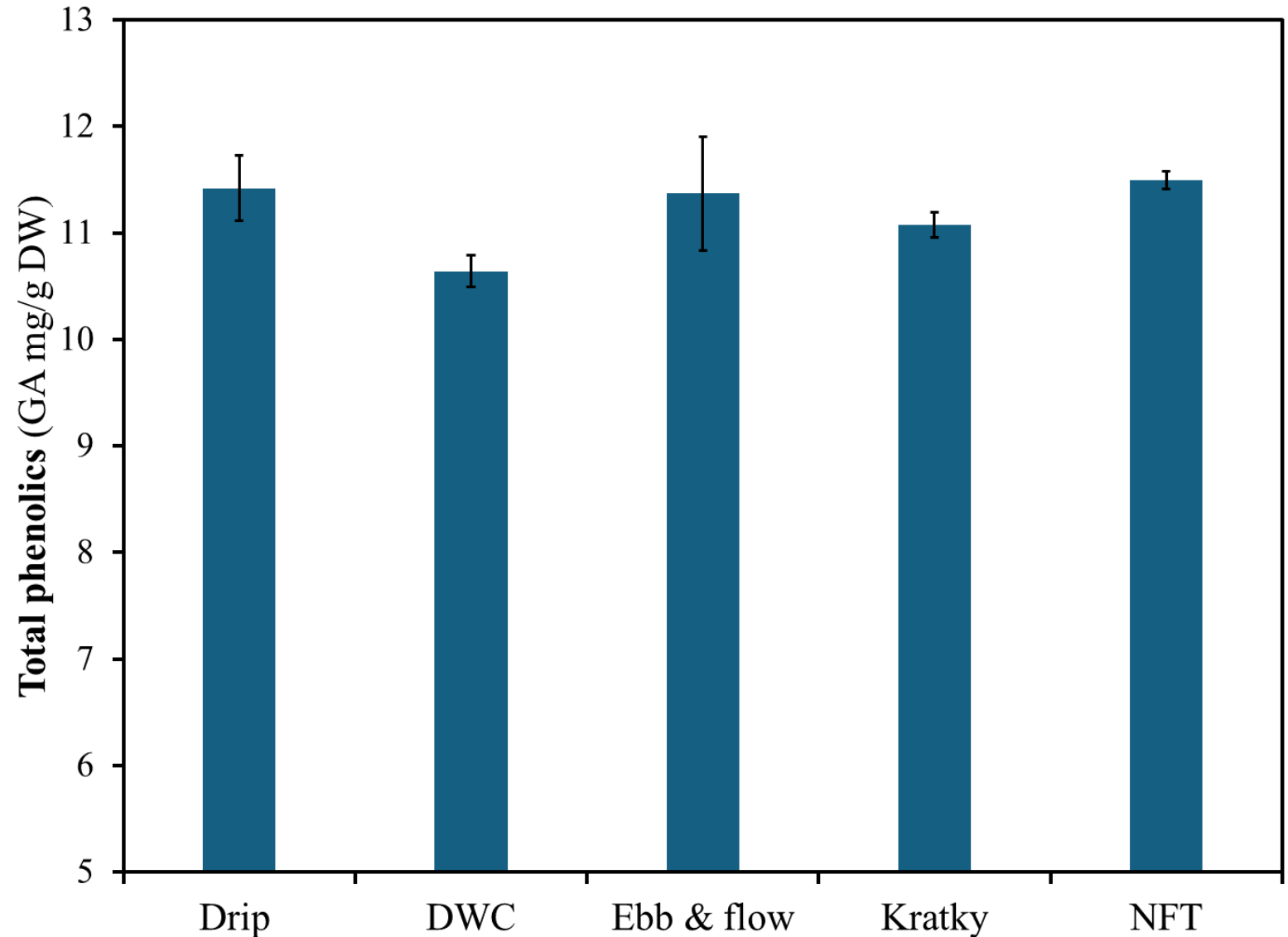
# What is the efficiency of indoor farming systems?

Impact on  
water use  
efficiency



# What is the efficiency of alternative soilless systems?

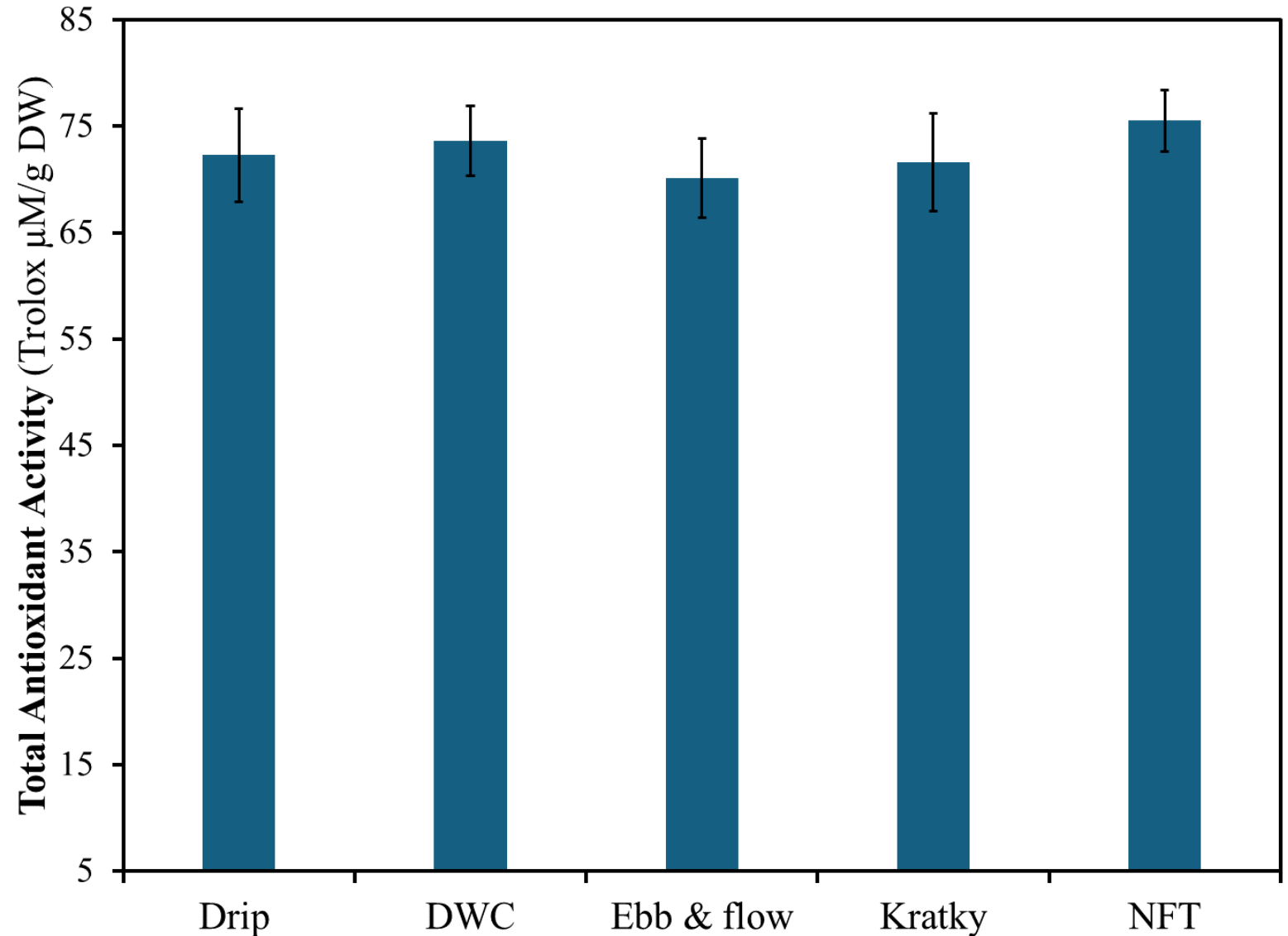
Impact on  
quality:  
Total  
phenolics  
content





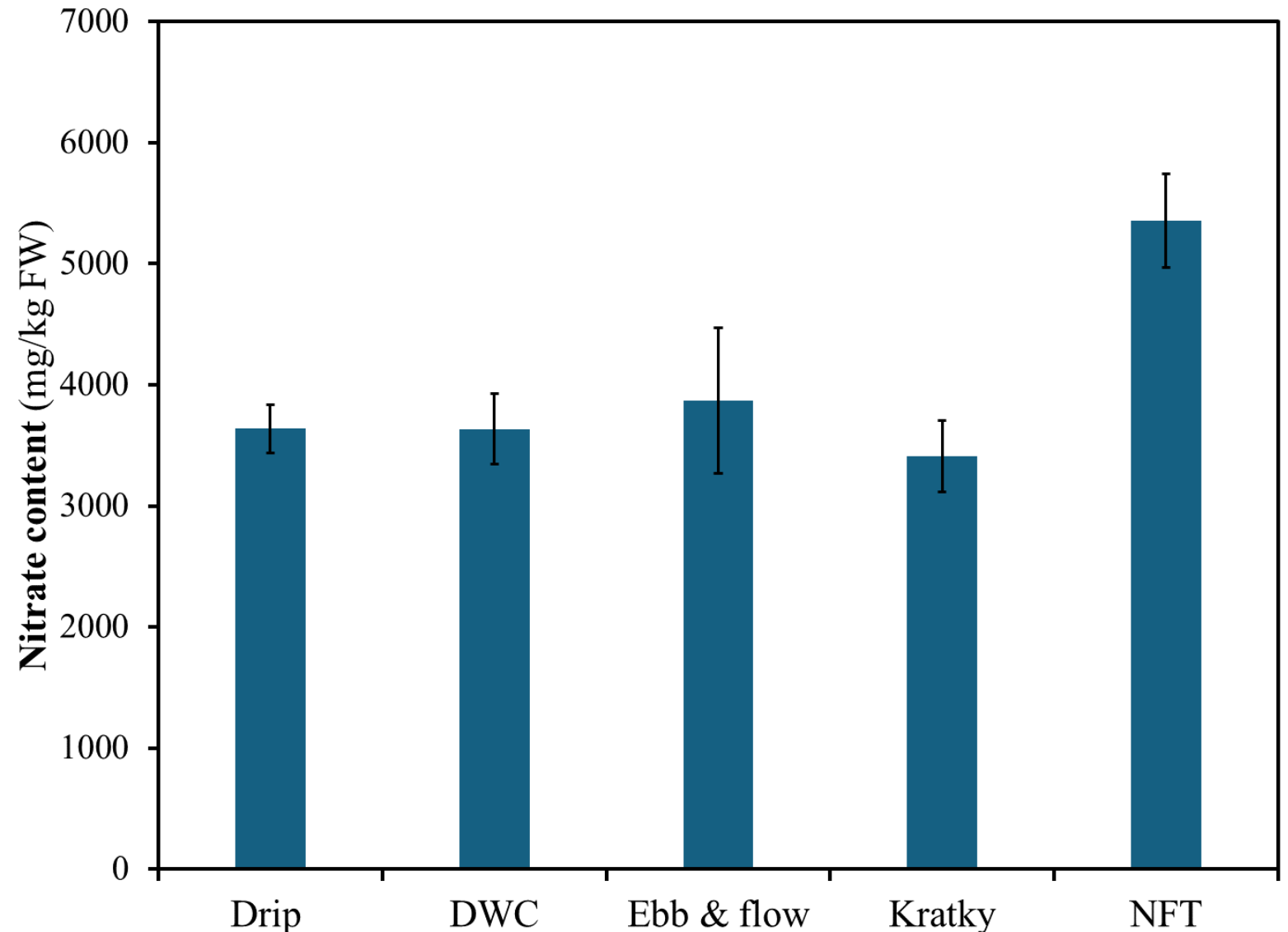
# What is the efficiency of alternative soilless systems?

Impact on  
quality:  
Total  
antioxidant  
activity



# What is the efficiency of alternative soilless systems?

Impact on  
quality:  
Nitrate  
content





# What is the efficiency of alternative soilless systems?

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## ➤ Which soilless growing system is more productive?

DWC > NFT > Kratky > Ebb&Flow > Drip

## ➤ Which one is more efficient?

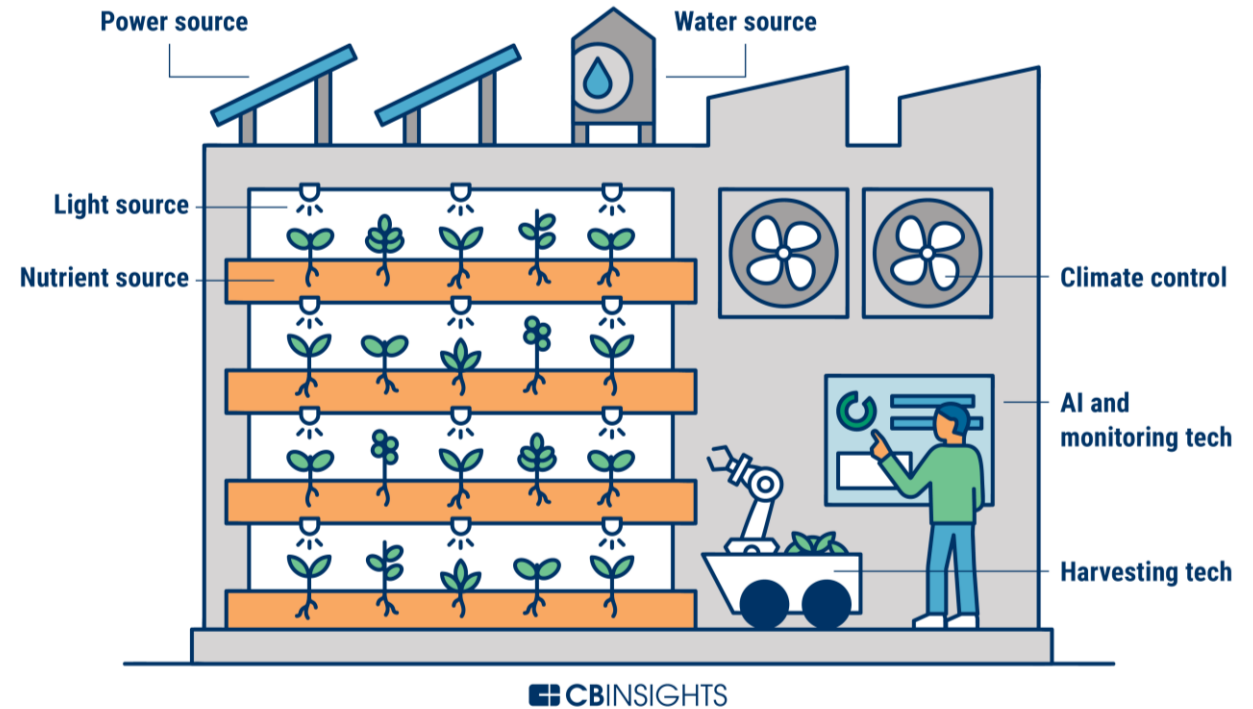
For water: DWC > Drip = Kratky = NFT > Ebb&Flow

For energy: Kratky = Drip > Ebb&Flow > NFT > DWC

## ➤ Which system is better for quality?

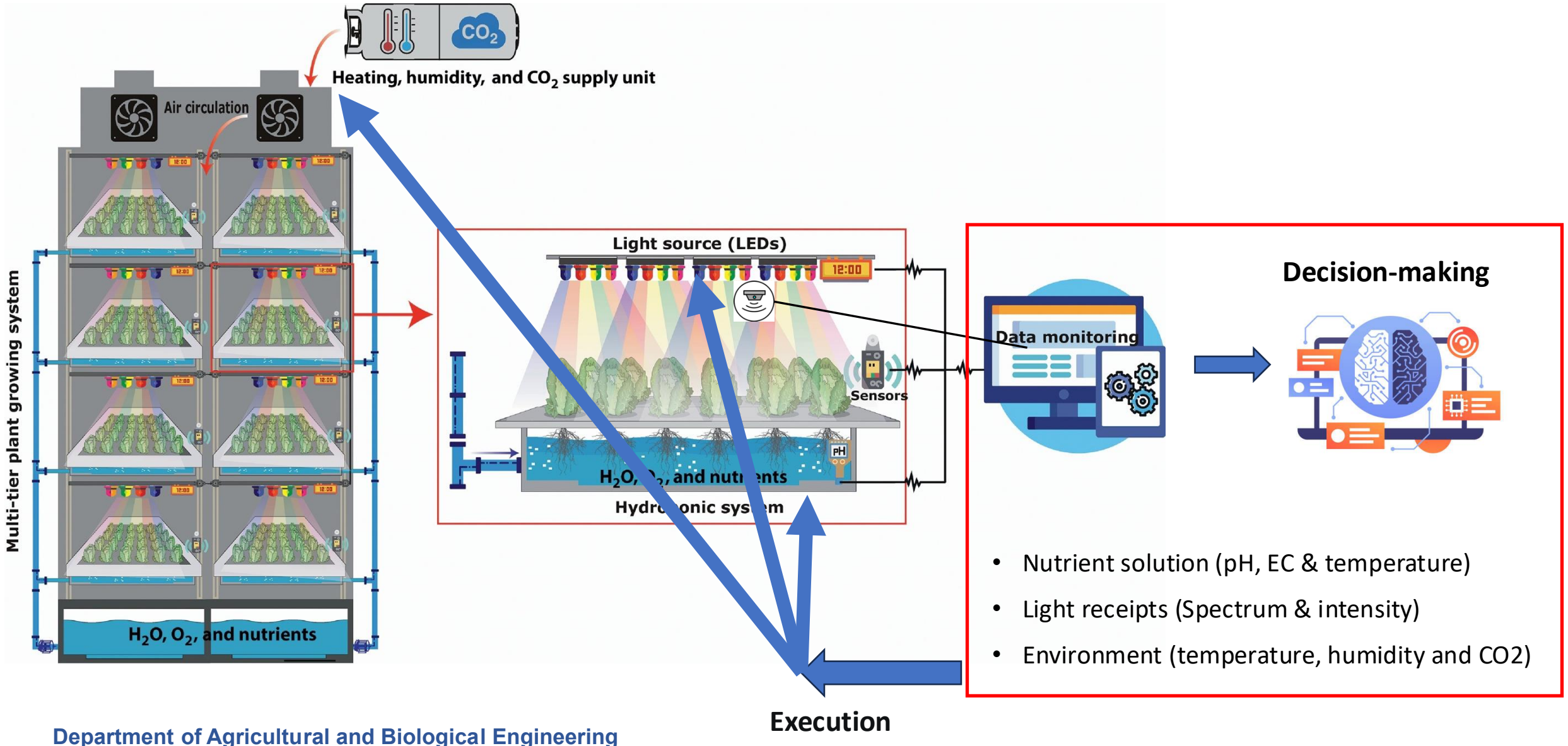
Kratky = Drip = Ebb&Flow > DWC=NFT

# The future of plant factory

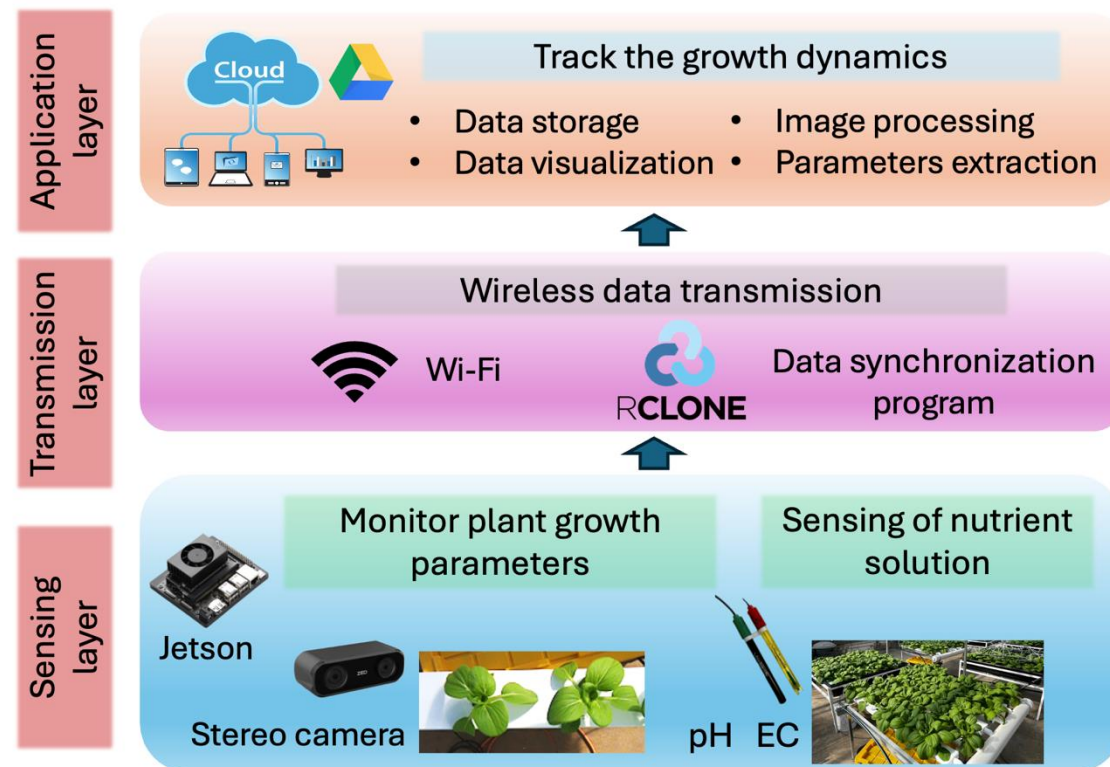




# The future of plant factory

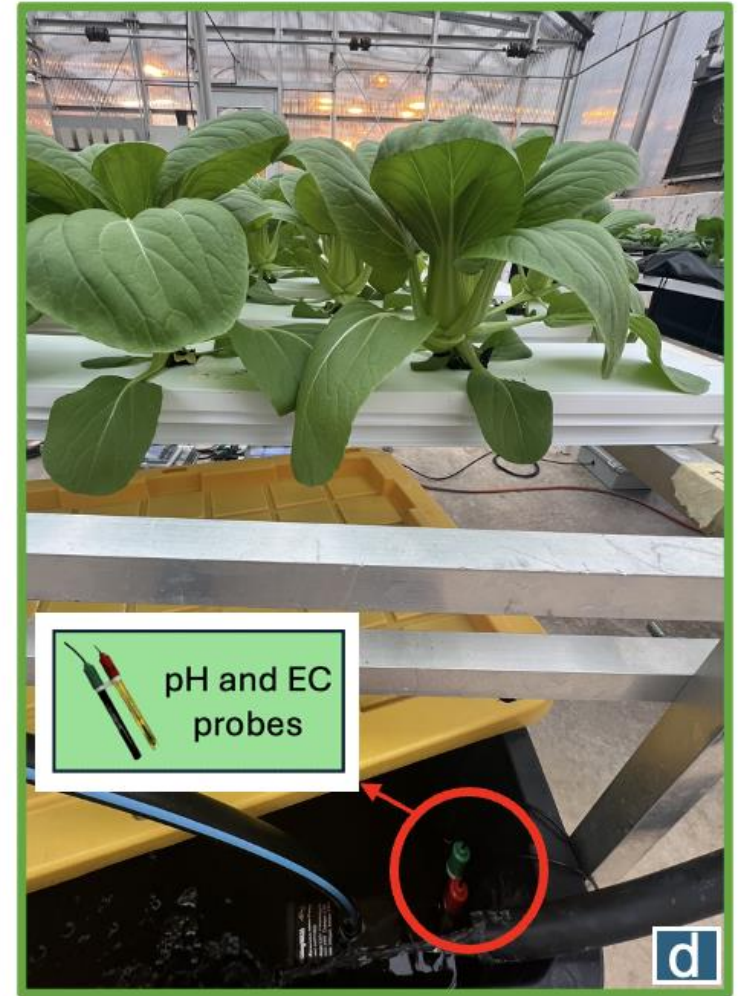
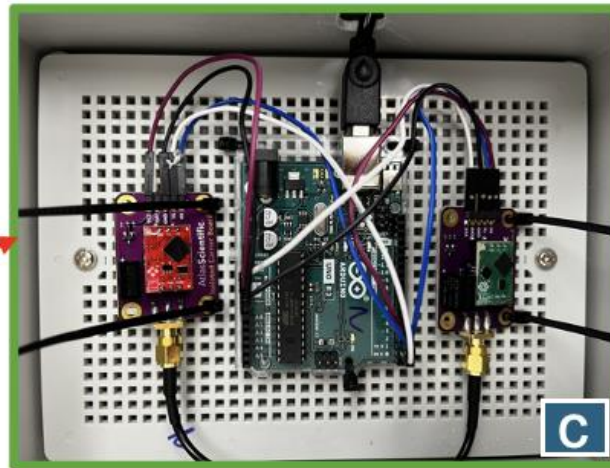


# Start with monitoring



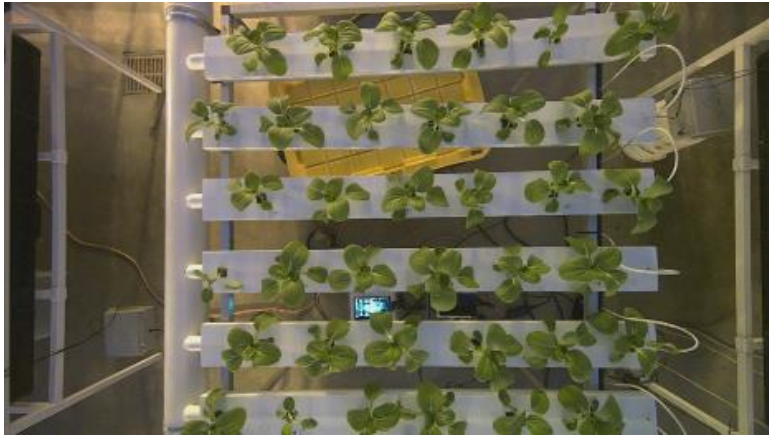


# An IoT-enabled vegetable growth monitoring system





# Segmentation is challenging!



Supervised learning



Labeling is tough 😞  
What if lettuce?



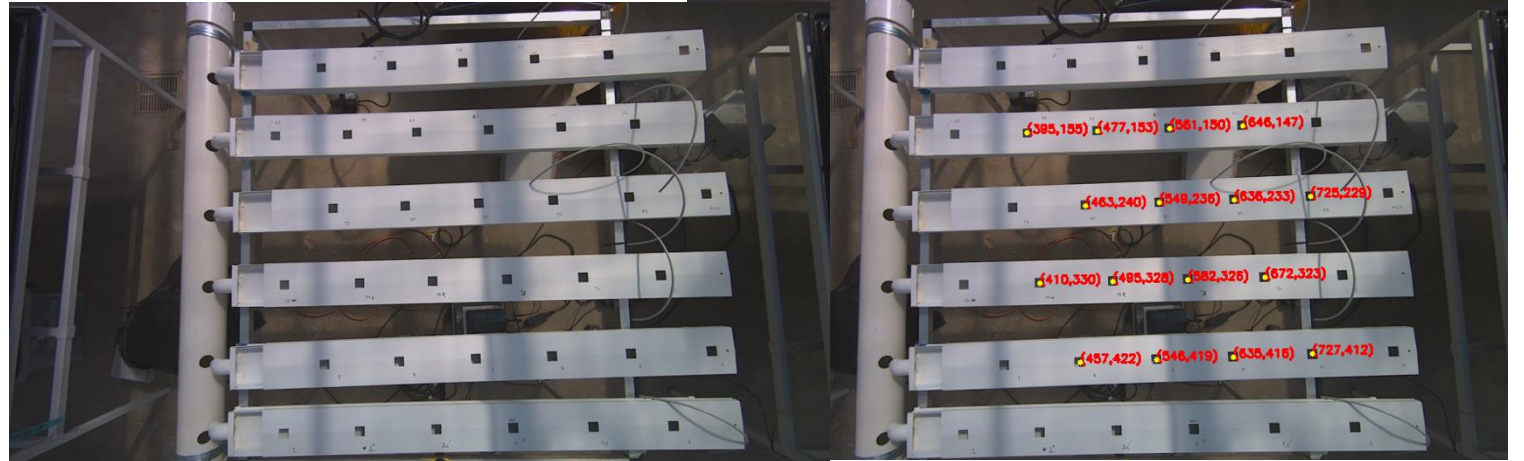
Unsupervised learning - [Segment Anything Model \(SAM\)](#)



Hey, we want individual  
plant first

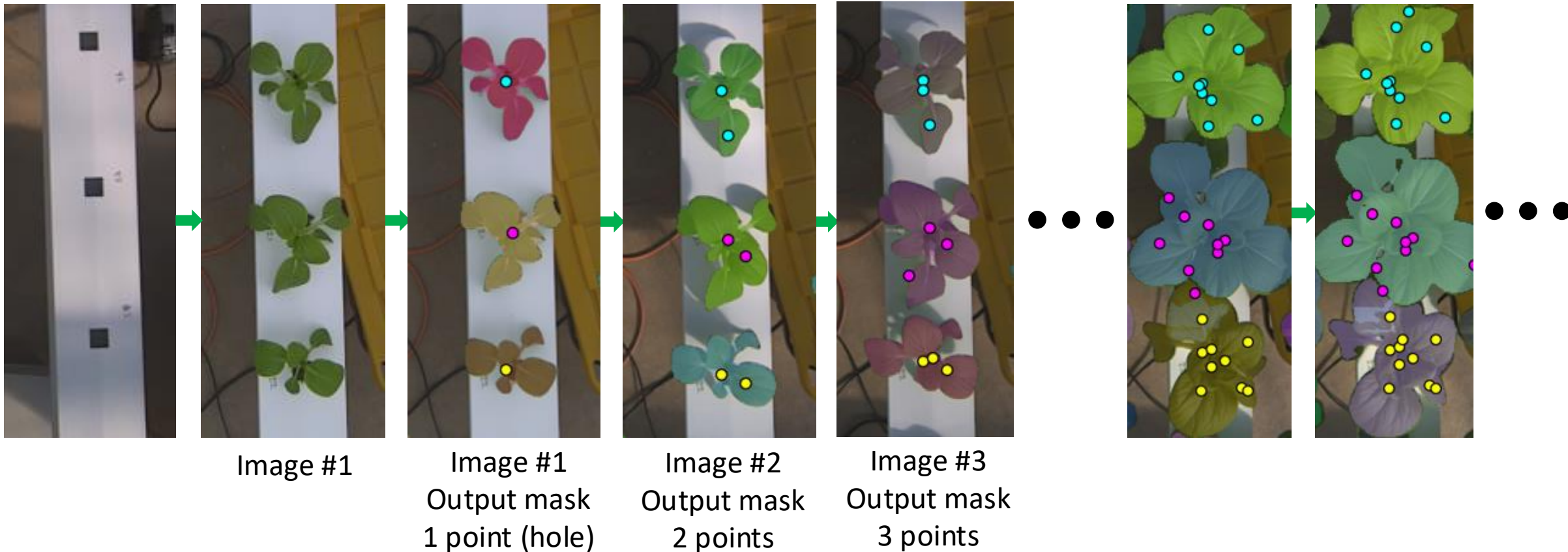


# SAM with single prompt points



# A recursive segmentation model (RSM)

## Recursive and sequential segmentation





# RSM



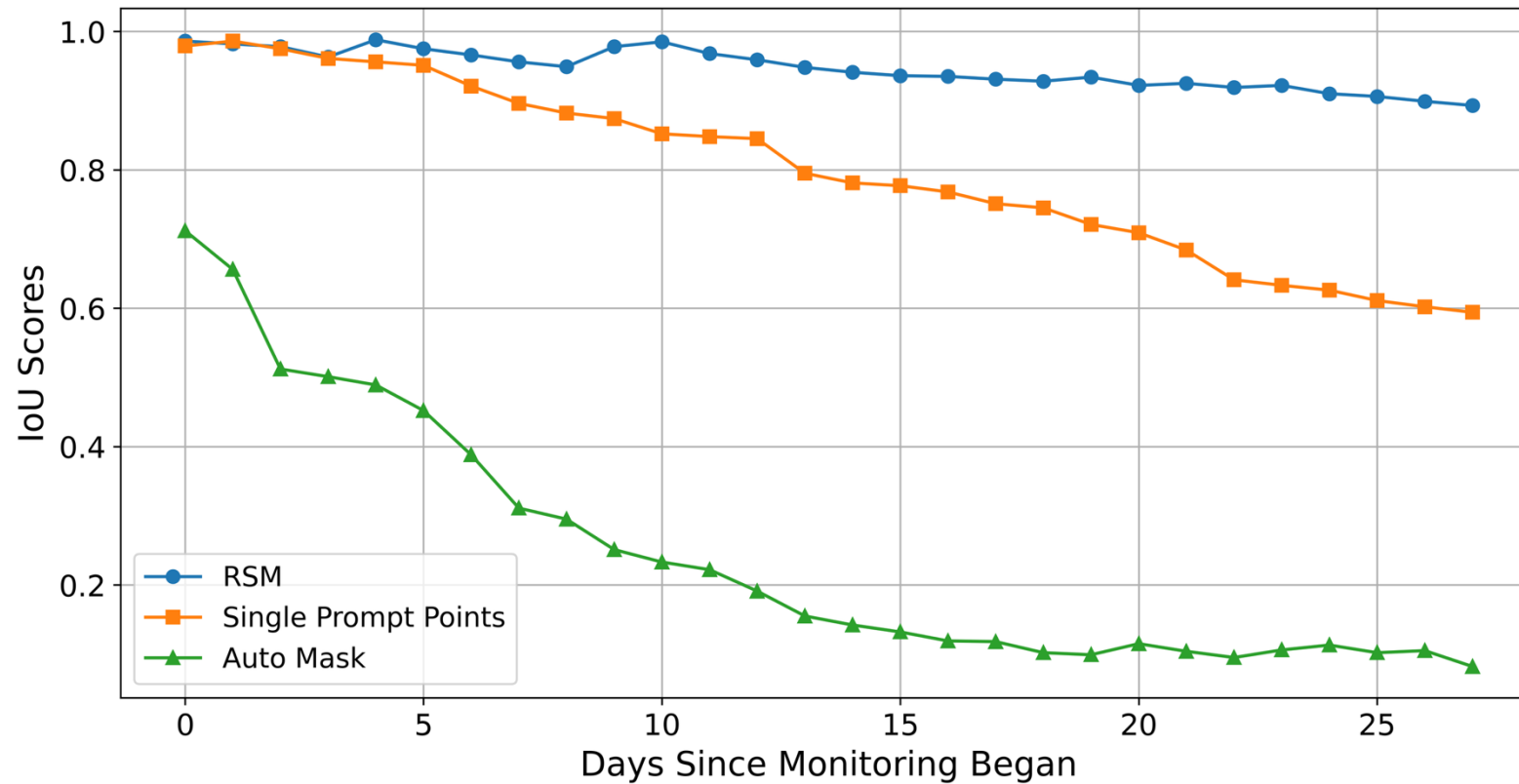
## Single Prompt Points



# Auto Mask



# Assessment with Intersection over Union (IoU)



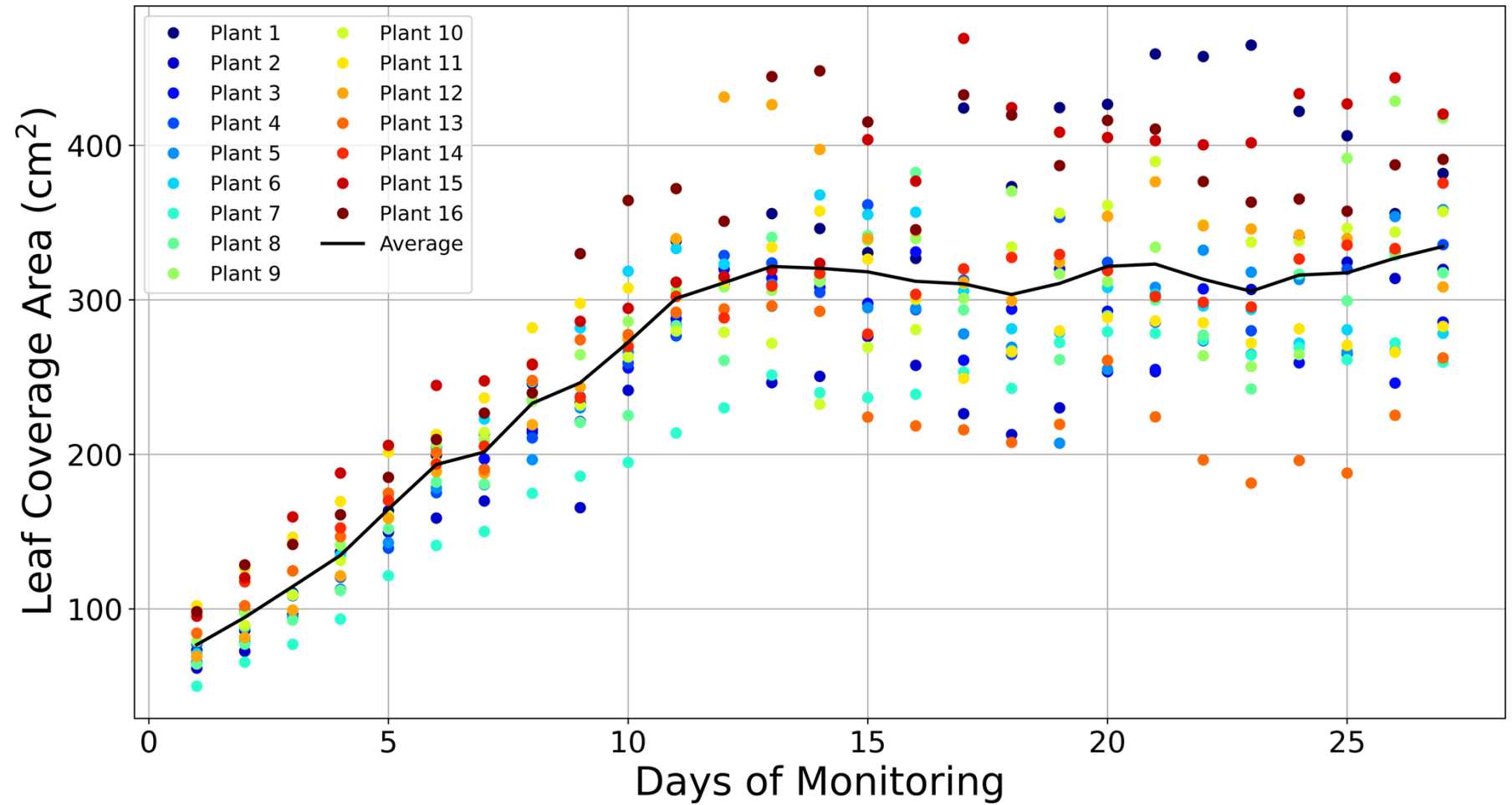
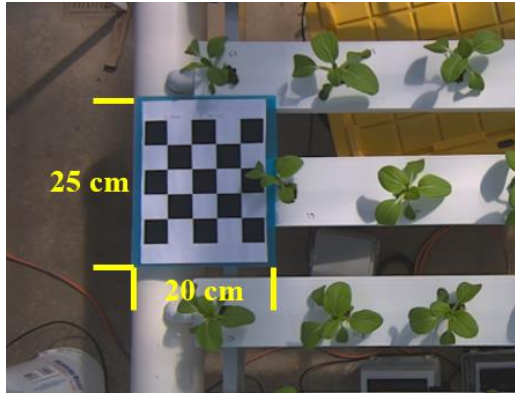
$$IoU = \frac{\text{Area of Intersection}}{\text{Area of Union}}$$

$$Avg IoU = \frac{\sum IoU}{16}$$

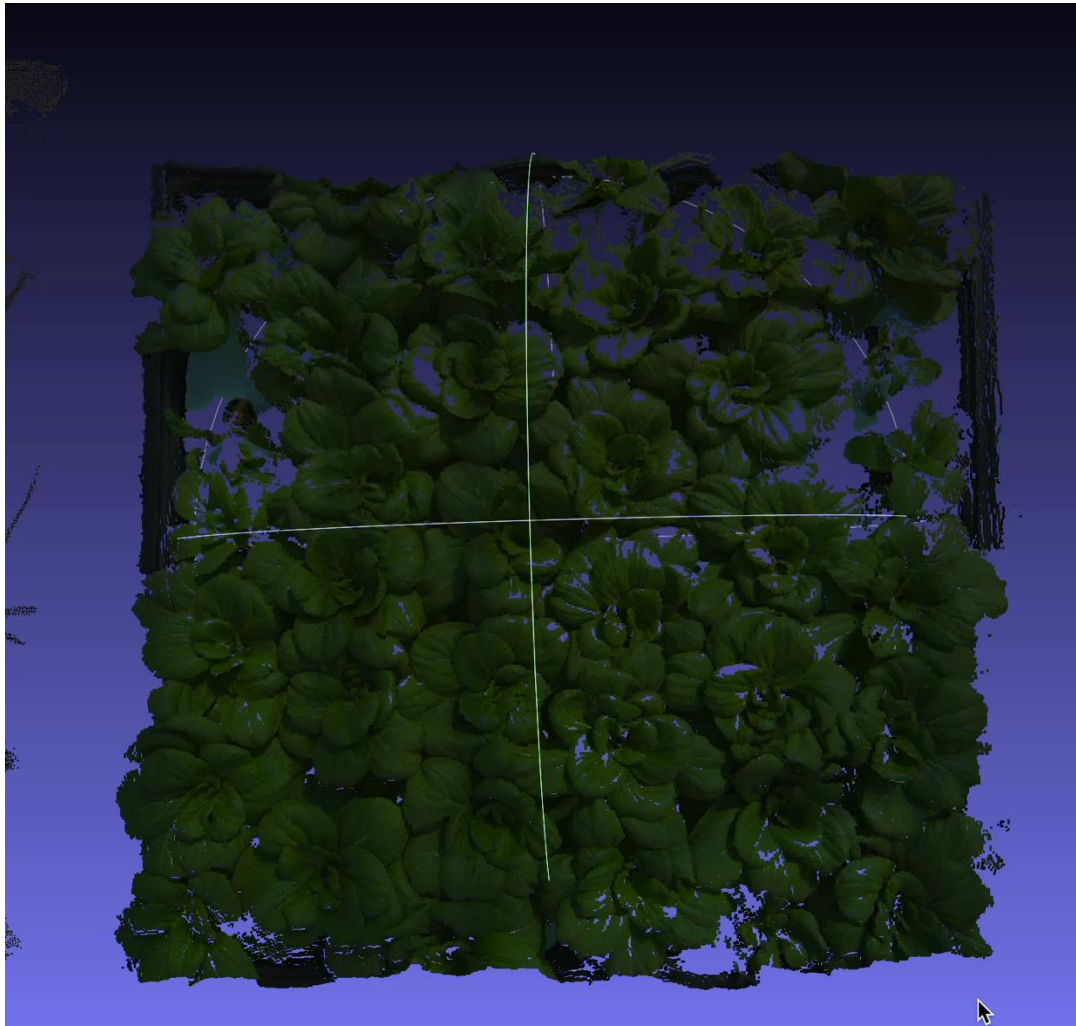




# Leaf Coverage Area Calculation and Tracking

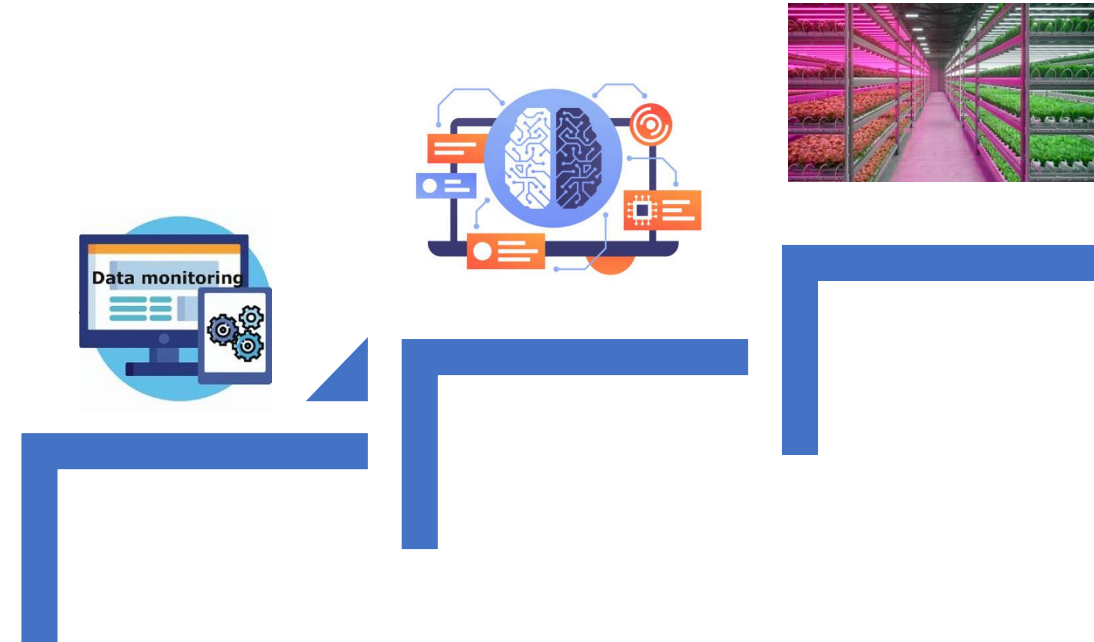
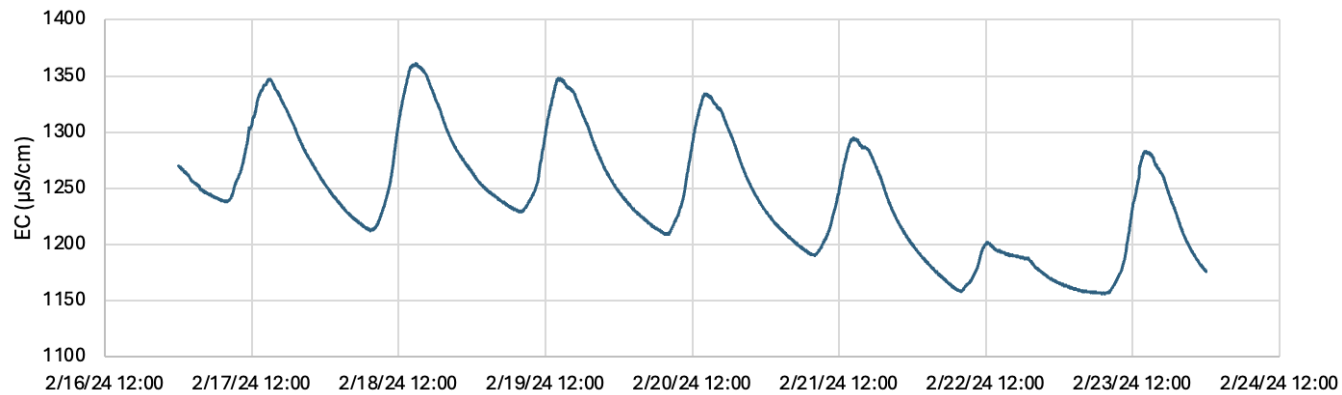


# 3D images (point clouds)





# Dynamic adjustments of nutrient management



# Acknowledgements



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United States Department of Agriculture (USDA)'s National Institute of Food and Agriculture (NIFA) Hatch Appropriations under Project **EN04822** and Accession No. **7005925** and under Project **PEN04723** and Accession No. **1020664**.

**Penn State Vegetable Crop Science Lab members:** Ben Morrison, Diane Lehr, Radhika Chalam, Pradip Poudel, Rishi Ravichandran, Trevor Johnson, Matthew Kaelin, Timothy Tretter ... and the **Greenhouse Manager:** Scott DiLoreto

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